Junghyun Jung, Ph.D.
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PROFESSIONAL AND ACADEMIC EXPERIENCE

- **Cedars-Sinai Medical Center** Los Angeles, CA, USA
  Research Data Scientist at Department of Computational Biomedicine *Jul. 2023 - present*
  Mentor: *Dr.* Kyoung Jae Won
  - **Development of foundation models for electronic health records**
    - Developed self-supervised transformer pretraining for cross-hospital representation learning.
    - Built a multi-institutional neonatal ICU data pipeline with harmonized vital-sign time series and documented clinical-concept provenance.
  - **Development of agentic AI and statistical methods**
    - Designed and deployed agentic AI for automated analysis and biological interpretation of spatial transcriptomics data.
    - Developed scalable AI-enabled pipelines integrating multi-modal omics, literature, and pharmacological databases for drug target prioritization and therapeutic hypothesis generation.
  - **Development of novel statistical methods for spatial single-cell genetics**
    - Created novel frameworks for detecting spatially-interacting eQTLs and region-specific gene regulation.
    - Applied methods to human brain and Alzheimer's disease datasets, revealing sex-specific genetic effects and spatial disease mechanisms.

  Project Data Manager at Department of Computational Biomedicine *Apr. 2026 - present*
  - **Data management and reproducibility infrastructure**
    - Responsible for FAIR data sharing compliance (CIRM DISC0O).
    - Responsible for building reproducible analysis pipelines (Nextflow, Docker/Apptainer, uv)

- **University of Southern California** Los Angeles, CA, USA
  Postdoctoral Researcher at Center for Genetic Epidemiology, Keck School of Medicine *Aug. 2020 - Jun. 2023*
  Mentor: *Dr.* Nicholas Mancuso
  - **Large-scale integrative analysis of complex disease**
    - Conducted multi-omic association studies integrating GWAS with multi-tissue QTL datasets (expression, splicing, enhancer, protein) to identify novel disease susceptibility genes and regulatory mechanisms.
    - Analyzed scRNA-seq, scATAC-seq, and TCR-seq data to characterize immune cell states, clonotype diversity, and clonal expansion.
  - **Characterized evolutionary constraints in disease genetics**
    - Inferred negative selection and polygenicity across molecular traits to understand disease architecture.

  Postdoctoral Researcher at Department of Clinical Pharmacy, School of Pharmacy *Aug. 2020 - Jul. 2021*
  Mentor: *Dr.* Serghei Mangul
  - **Identified genetic drivers of COVID-19 outcomes**
    - Analyzed WES data to discover rare variants in innate immunity pathways influencing disease severity.
    - Performed GWAS across multi-ancestry to identify common risk variants.
  - **Benchmarked HLA typing methods for RNA-seq**
    - Evaluated multiple HLA callers for accuracy, efficiency, and sequencing parameter effects.
    - Developed consensus HLA typing approaches and assessed cross-ancestry performance disparities.

- **Dongguk University** Seoul, Korea
  Postdoctoral Researcher at Department of Computer Science and Engineering *Mar. 2020 - Jul. 2020*
  Mentor: *Dr.* Jong Wha J. Joo
  - **Set-based analysis to increase the power of association testing**
    - Improved association power using allelic heterogeneity and variant aggregation
    - Built high-dimensional multi-phenotype models correcting for population structure

- **Dongguk University** Seoul, Korea
  Instructor at Convergence Software Institute Dongguk *Mar. 2018 - Aug. 2020*
  (Biological Information Software Convergence)

- **Dongguk University** Seoul, Korea
  Teaching assistant at Department of Life Science *Sept. 2012 - Aug. 2013*

EDUCATION

- **Ph.D. in Life Science** | Dongguk University, Seoul, Korea | *Sept. 2014 – Feb. 2020*
  Dissertation project: Bioinformatics-based approaches for identifying novel genetic markers

- **M.S. in Life Science** | Dongguk University, Seoul, Korea | *Mar. 2012 – Aug. 2014*
  Dissertation project: Studies on the *in silico* analyzing methods related to toxic effects and mechanisms of chemicals

- **B.S. in Life Science** | Dongguk University, Seoul, Korea | *Mar. 2007 – Feb. 2012*

PREPRINT AND UNDER REVIEW

- G. Weng, D. Zhu, Y. Zhao, PC. Martin, J. Jung, et al.,, Cross-domain confidence reliability and remappability of frozen single-cell representations, *Genome Biology*, under review

- J. Jung and KJ. Won, Degree-GEAR — Degree-Preserving Graph Evidence Aggregation and Reweighting for Spatial Domain Identification, *AAAI-27*, under review

- J. Jung, T. Parimon, et al.,, DeepMAST: An Evidence-Governed Agent Harness for Literature-Grounded Interpretation and Drug-Target Prioritization in Spatial Transcriptomics, under review

- H. Choi, H. Kim, J. Jung, et al.,, SPIX: Superpixel-based multiscale framework for scalable spatial transcriptomics analysis, *Science Advances*, under review

- N. Cho, B. Park, GY. Lee, J. Jung et al.,, PersonaAI: An Interactive Agentic-AI Framework for Autonomous Hypothesis Generation and Validation in Aging, *Nature Aging*, under review

- H. Kim, B. Park, J. Jung, et al.,, Identifying Context-Specific Cell-Cell Interaction Genes Without Ligand-Receptor Databases from Spatial Transcriptomics, *Nature Communications*, under review

- J. Song, J. Jung, et al.,, Leveraging genetic effects in Graves' disease via cell type-specific network diffusion, *Communications Biology*, under review

- D. Yu and R. Ayyala, J. Jung, et al.,, A rigorous benchmarking of alignment-based HLA typing algorithms for RNA-seq data, *bioRxiv*, 2023.05.22.541750

PUBLICATIONS

[1] W. Oh, **J. Jung**, and J. W. J. Joo, " MR-GGI: accurate inference of gene–gene interactions using Mendelian randomization," *BMC Bioinformatics*, vol. 25, no. 1, pp. 1–16, 2024.

[2] J. Song, D. Kim, **J. Jung**, E. Choi, *et al.*, "Elucidating immunological characteristics of the adenoma-carcinoma sequence in colorectal cancer patients in South Korea using a bioinformatics approach," *Scientific reports*, vol. 14, no. 10105, 2024.

[3] D. Kim, J. Song, S. Mangul, N. Mancuso, **J. Jung***, and W. Jang*, "Large-scale integrative analysis of juvenile idiopathic arthritis for new insight into its pathogenesis," *Arthritis Research & Therapy*, vol. 26, no. 1, p. 47, 2024.

[4] Z. Zhang, **J. Jung**, A. Kim, N. Suboc, S. Gazal, and N. Mancuso, "A scalable approach to characterize pleiotropy across thousands of human diseases and complex traits using GWAS summary statistics," *The American Journal of Human Genetics*, vol. 110, no. 11, pp. 1863–1874, 2023.

[5] **J. Jung***, Z. Lu, A. de Smith, and N. Mancuso*, "Novel insight into the etiology of ischemic stroke gained by integrative multiome-wide association study," *Human Molecular Genetics*, vol. 33, no. 2, pp. 170–181, 2024.

[6] G. Butler-Laporte, G. Povysil, J. A. Kosmicki, E. T. Cirulli, T. Drivas, S. Furini, C. Saad, A. Schmidt, P. Olszewski, U. Korotko, **J. Jung**, *et al.*, "Exome-wide association study to identify rare variants influencing COVID-19 outcomes: Results from the Host Genetics Initiative," *PLoS Genetics*, vol. 18, no. 11, p. e1010367, 2022.

[7] W. group lead Andrews Shea J. 6 Kanai Masahiro 3 Cordioli Mattia 7, M. analyses team lead, M. analyses team members: PHEWAS Polimanti Renato 1, M. analyses team members: Mendelian randomization Harerimana Nadia 8, M. analyses team members: methods development Pirinen Matti 7, g. p. Manuscript analyses team members: PC projection, P. management lead Liao Rachel G. 3, P. managment support Chwialkowska Karolina 9 Trankiem Amy 3 Balaconis Mary K. 3, H. Nguyen, *et al.*, "A first update on mapping the human genetic architecture of COVID-19," *Nature*, vol. 608, no. 7921, pp. E1–E10, 2022.

[8] Y. Shim, M. Shin, **J. Jung**, B. Koo, and W. Jang, "An in-silico approach to studying a very rare neurodegenerative disease using a disease with higher prevalence with shared pathways and genes: Cerebral adrenoleukodystrophy and Alzheimer's disease," *Frontiers in molecular neuroscience*, vol. 15, 2022.

[9] J. Song, D. Kim, S. Lee, **J. Jung**, J. W. J. Joo, and W. Jang, "Integrative transcriptome-wide analysis of atopic dermatitis for drug repositioning," *Communications biology*, vol. 5, no. 1, pp. 1–13, 2022.

[10] J. L. Wiemels, R. Wang, M. Zhou, H. Hansen, R. Gallant, **J. Jung**, N. Mancuso, A. J. De Smith, C. Metayer, S. C. Kogan, *et al.*, "Cytomegalovirus proteins, maternal pregnancy cytokines, and their impact on neonatal immune cytokine profiles and acute lymphoblastic leukemogenesis in children," *Haematologica*, 2022.

[11] A. Kousathanas, E. Pairo-Castineira, K. Rawlik, A. Stuckey, C. A. Odhams, S. Walker, C. D. Russell, T. Malinauskas, Y. Wu, J. Millar, **J. Jung**, *et al.*, "Whole genome sequencing reveals host factors underlying critical Covid-19," *Nature*, pp. 1–10, 2022.

[12] C. Fallerini, N. Picchiotti, M. Baldassarri, K. Zguro, S. Daga, F. Fava, E. Benetti, S. Amitrano, M. Bruttini, M. Palmieri, **J. Jung**, *et al.*, "Common, low-frequency, rare, and ultra-rare coding variants contribute to COVID-19 severity," *Human genetics*, pp. 1–27, 2021.

[13] M. D'Antonio, J. P. Nguyen, T. D. Arthur, H. Matsui, B. M. Neale, M. Daly, A. Ganna, C. Stevens, G. A. Pathak, S. J. Andrews, **J. Jung**, *et al.*, "SARS-CoV-2 susceptibility and COVID-19 disease severity are associated with genetic variants affecting gene expression in a variety of tissues," *Cell reports*, vol. 37, no. 7, p. 110020, 2021.

[14] B. Lee, M. K. Shin, I.-W. Hwang, **J. Jung**, Y. J. Shim, G. W. Kim, S. T. Kim, W. Jang, and J.-S. Sung, "A Deep Learning Approach with Data Augmentation to Predict Novel Spider Neurotoxic Peptides," *International Journal of Molecular Sciences*, vol. 22, no. 22, p. 12291, 2021.

[15] G. Povysil, G. Butler-Laporte, N. Shang, C. Wang, A. Khan, M. Alaamery, T. Nakanishi, S. Zhou, V. Forgetta, R. J. Eveleigh, **J. Jung**, *et al.*, "Rare loss-of-function variants in type I IFN immunity genes are not associated with severe COVID-19," *The Journal of Clinical Investigation*, vol. 131, no. 14, 2021.

[16] F. Hormozdiari†, **J. Jung**†, E. Eskin, and J. W. J. Joo, "MARS: leveraging allelic heterogeneity to increase power of association testing," *Genome biology*, vol. 22, pp. 1–26, April 2021.

[17] D. Kim, J. Song, S. Lee, **J. Jung**, and W. Jang, "An Integrative Transcriptomic Analysis of Systemic Juvenile Idiopathic Arthritis for Identifying Potential Genetic Markers and Drug Candidates," *International Journal of Molecular Sciences*, vol. 22, no. 2, p. 712, 2021.

[18] J. H. Choi, T. Kim, **J. Jung**, and J. W. J. Joo, "Fully automated web-based tool for identifying regulatory hotspots," *BMC genomics*, vol. 21, no. 10, pp. 1–7, 2020.

[19] G. J. Lee, S. M. Park, **J. Jung**, and J. W. J. Joo, "A Fully Automated Parallel-Processing R Package for High-Dimensional Multiple-Phenotype Analysis Considering Population Structure," *International Journal of Fuzzy Logic and Intelligent Systems*, vol. 20, no. 3, pp. 219–226, 2020.

[20] J. Song, D. Kim, J. Hong, G. W. Kim, **J. Jung**, S. Park, H. J. Park, J. W. J. Joo, and W. Jang, "Meta-analysis of polymyositis and dermatomyositis microarray data reveals novel genetic biomarkers," *Genes*, vol. 10, no. 11, p. 864, 2019.

[21] **J. Jung**, G. W. Kim, B. Lee, J. W. J. Joo, and W. Jang, "Integrative genomic and transcriptomic analysis of genetic markers in Dupuytren's disease," *BMC medical genomics*, vol. 12, no. 5, pp. 1–10, 2019.

[22] E. Lee, H. Jeon, M. Lee, J. Ryu, C. Kang, S. Kim, **J. Jung**, and Y. Kwon, "Molecular origin of AuNPs-induced cytotoxicity and mechanistic study," *Scientific reports*, vol. 9, no. 1, pp. 1–13, 2019.

[23] **J. Jung**, G. W. Kim, W. Lee, C. Mok, S. H. Chung, and W. Jang, "Meta-and cross-species analyses of insulin resistance based on gene expression datasets in human white adipose tissues," *Scientific reports*, vol. 8, no. 1, pp. 1–13, 2018.

[24] H. Kim, J. Yoo, J. Shin, Y. Chang, **J. Jung**, D.-G. Jo, J. Kim, W. Jang, C. J. Lengner, B.-S. Kim, *et al.*, "Modelling APOE 3/4 allele-associated sporadic Alzheimer's disease in an induced neuron," *Brain*, vol. 140, no. 8, pp. 2193–2209, 2017.

[25] J. Yoo, E. Lee, H. Y. Kim, D.-H. Youn, **J. Jung**, H. Kim, Y. Chang, W. Lee, J. Shin, S. Baek, *et al.*, "Electromagnetized gold nanoparticles mediate direct lineage reprogramming into induced dopamine neurons in vivo for Parkinson's disease therapy," *Nature nanotechnology*, vol. 12, no. 10, pp. 1006–1014, 2017.

[26] **J. Jung**, K. Hah, W. Lee, and W. Jang, "Meta-analysis of microarray datasets for the risk assessment of coplanar polychlorinated biphenyl 77 (PCB77) on human health," *Toxicology and Environmental Health Sciences*, vol. 9, no. 2, pp. 161–168, 2017.

[27] **J. Jung**, C. Mok, W. Lee, and W. Jang, "Meta-analysis of microarray and RNA-Seq gene expression datasets for carcinogenic risk: An assessment of Bisphenol A," *Molecular & Cellular Toxicology*, vol. 13, no. 2, pp. 239–249, 2017.

[28] E. Lee, **J. Jung**, D. Jung, C. S. Mok, H. Jeon, C.-S. Park, W. Jang, and Y. Kwon, "Inhibitory effects of novel SphK2 inhibitors on migration of cancer cells," *Anti-Cancer Agents in Medicinal Chemistry (Formerly Current Medicinal Chemistry-Anti-Cancer Agents)*, vol. 17, no. 12, pp. 1689–1697, 2017.

[29] H. G. Kang, D.-H. Kim, S.-J. Kim, Y. Cho, **J. Jung**, W. Jang, and K.-H. Chun, "Galectin-3 supports stemness in ovarian cancer stem cells by activation of the Notch1 intracellular domain," *Oncotarget*, vol. 7, no. 42, p. 68229, 2016.

Conference Presentations - Oral presentation

- Z. Zhang, **J. Jung**, N. Suboc, S. Gazal, N. Mancuso. A variational Bayesian approach to characterize pleiotropic components across thousands of human diseases and complex traits using GWAS summary statistics. ASHG, Los Angeles, CA, USA, 2022
- **J. Jung**, G.W. Kim, B. Lee, J.W.J. Joo., W. Jang. Integrative genomic and transcriptomic analysis for genetic markers in Dupuytren's disease. TBC/BIOINFO 2018. Seoul, Korea, 2018
- **J. Jung**, G.W. Kim, J.H. Park, S.H. Chung, W. Jang . Meta-analysis of insulin resistance based on gene expression datasets in human adipose tissues. The 13th international congress on obesity (ICO). Vancouver, Canada, 2016.
- G.W. Kim, **J. Jung**, H.K. Jo, K.J. Jeong, W. Jang, S.H. Chung. Elucidation of suppressive mechanism of betulinic acid on adipogenesis using global gene expression profiling. The 13th international congress on obesity (ICO). Vancouver, Canada, 2016.

Conference Presentations - Poster presentation

- **J. Jung**[†], S. Lee[†], J. Song, C. Wang, W. Jang[*], k.J. Won[*]. A Multimodal Framework Integrating Spatial Transcriptomics and GWAS Identifies Spatially Resolved Genetic Mechanisms in Alzheimer's Disease. ASHG, Boston, USA, 2025
- **J. Jung**[†], J. Song[†], C. Wang, W. Jang[*], k.J. Won[*]. Unraveling spatially-interacting genetic variants in the human dorsolateral prefrontal cortex. RECOMB, Seoul, Korea, 2025
- **J. Jung**[†], S. Lee[†], J. Song, C. Wang, W. Jang[*], k.J. Won[*]. REMIX: Identifying Region-specific Spatially Variable Genes Associated with Alzheimer's Disease Progression. RECOMB, Seoul, Korea, 2025
- Z. Lu, J. Landau, **J. Jung**, D. Yuan, S. Gazal, N. Mancuso. Multi-ancestry fine-mapping of gene expression data identifies shared cis-regulatory mechanisms. Probabilistic Modeling in Genomics (ProbGen), Cold Spring Harbor, USA, 2023
- Z. Zhang, **J. Jung**, S. Gazal, N. Mancuso. A variational Bayesian approach to characterize pleiotropic components across thousands of human diseases and complex traits using GWAS summary statistics. Probabilistic Modeling in Genomics (ProbGen), Cold Spring Harbor, USA, 2023
- **J. Jung**, Z. Lu, N. Mancuso. Novel insight into the etiology of ischemic stroke gained by integrating human transcriptome-wide association study with rodent expression data. ASHG, Virtual, 2021
- Z. Zhang, **J. Jung**, N. Suboc, S. Gazal, N. Mancuso. A variational Bayesian approach to characterize latent genetic components using GWAS summary statistics. ASHG, Virtual, 2021
- Z. Lu, **J. Jung**, S. Gazal, N. Mancuso. Residual proteome-wide association study identifies genes for blood-related traits. ASHG, Virtual, 2021
- **J. Jung**, R. Brown, W. Jang, B. Pasaniuc, J.W.J. Joo and E. Eskin. GGmend: A Mendelian randomization method for finding gene-on-gene regulatory effects in the presence of unobserved confounders. 23th RECOMB, Washington, D.C., USA, 2019.
- F. Hormozdiari, **J. Jung**, E. Eskin and J.W.J. Joo. Leveraging allelic heterogeneity to increase the power of association testing. 23th RECOMB, Washington, D.C., USA, 2019.
- J. Song, W. Jang, J. Jung, D.E. Kim and J.Y. Hong W. Jang. Meta-analysis of polymyositis and dermatomyositis microarray data sets. 23th RECOMB, Washington, D.C., USA, 2019.
- **J. Jung**, G.W. Kim, J.H. Park, S.H. Chung, W. Jang. Cross-species analysis reveals molecular mechanisms of insulin resistance in white adipose tissues. 12th International Conference on Toxicogenomics (ICT), Seoul, Korea, 2016.
- **J. Jung**, G.W. Kim, J.H. Park, S.H. Chung, W. Jang. Integrative cross-species analysis of insulin resistance expression profile in white adipose tissues. 9th International Conference on Environmental Health Sciences (ICoEHS), Incheon, Korea, 2016.

- **J. Jung**, J.H. Park, W. Jang. Meta-analysis of microarray gene expression datasets for silver nanoparticles. 11th International Conference on Toxicogenomics (ICT), Seoul, Korea, 2015.
- **J. Jung**, J.H. Park, W. Jang. Meta-analysis of microarray and RNA-Seq gene expression datasets for carcinogenic risk assessment of Bisphenol A. 8th International Conference on Environmental Health Sciences (ICoEHS), Incheon, Korea, 2015.
- **J. Jung**, J.H. Park, W. Jang. Meta-analysis of microarray gene expression datasets for in silico risk assessment of Bisphenol A. 6th International Conference on Environmental Health Sciences (ICoEHS), Incheon, Korea, 2013.

PATENTS

- Genetic markers for diagnosing JIA and diagnosis method using these markers. 2024.12.24. Korea (1027481570000)
- Composition for diagnosing Insulin Resistance and its uses. 2020.04.10. Korea (1021018070000)
- Diagnostic kit comprising biomarkers for bisphenol A exposure diagnosis. 2018.12.21. Korea (1019336950000)
- Diagnostic kit comprising biomarkers for genistein exposure diagnosis. 2018.12.21. Korea (1019336940000)
- Diagnostic kit comprising biomarkers for polychlorinated biphenyl 77 exposure diagnosis. 2018.12.21. Korea (1019336920000)

HONORS AND AWARDS

- **Sejong Science Fellowship (Overseas Training Track)** - The Ministry of Science and ICT, Korea, *2024*
- **Dongguk Global Researcher Scholarship**, *2014*
- **13th International Conference on Toxicogenomics (ICT)**, Best poster, The Korea Society of Toxicogenomics and Toxicoproteomics, *2016*
- **9th International Conference on Environmental Health Sciences (ICoEHS)**, Best poster, Society of Environmental Risk Assessment and Health Science, *2016*
- **6th International Conference on Environmental Health Sciences (ICoEHS)**, Best poster, Society of Environmental Risk Assessment and Health Science, *2013*

TEACHING EXPERIENCE

- **Instructor for Big Data Processing/Analysis using Public Databases (BSI4002-01 and BSI4001-01, 3 credits)**
  Convergence Software Institute Dongguk (2018 - 2020)

  - **Big Data Processing Using Public Biological Databases (BSI4002-01)**: Delivered a 16-week undergraduate course on biological big data processing and analysis from public database using Python and R, with emphasis on equipping students with practical computational skills directly applicable to biomedical research and the life science industry.
  - **Big Data Analysis Using Public Biological Databases (BSI4001-01)**: Delivered a 16-week undergraduate course covering Python programming and R-based biological data analysis, including sequence/variant analysis, transcriptome analysis, gene ontology, and pathway analysis, with an equal split of lecture and hands-on computational practice.

- **Invited lecturer on Statistical Methods in Human Genetics (PM 570)**
  University of Southern California (spring 2023)

  I introduce cutting-edge tools for genome-wide association and genetic perturbations analysis that have been recently published.

- **Workshop on Genomic Data Analysis and Deep Learning**
  LINC+ Donguk University (Winter 2019)

  - A 1-month workshop introducing attendees to the basics of Omics data analysis. Topics include preprocessing of RNA-seq and expression microarray data, differentially expressed genes (DEGs) analysis and functional enrichment analysis.

PROFESSIONAL SERVICE

- **Editorial Board Member**: BMC Genomics (May 2025 -)
- **Review Editor**: Frontiers in Molecular Bioscience (June 2023 -)
- **Reviews**: Genome medicine, The American Journal of Human Genetics, Translational Stroke Research, Cardiovascular Diabetology, BMC Medicine, BMC Genomics, Molecular Neurobiology, European Journal of Medical Research, Scientific Reports

RESEARCH TECHNIQUE

- **Computational Skills**
  - **AI Agent Frameworks**: LangChain/LangGraph, Google Agent Development Kit (ADK)
  - **Agent Communication Protocols**: MCP, A2A Protocol
  - **NGS Data Analysis**: WGS, WES, RNA-seq, ATAC-seq, ChIP-seq, and scRNA/scATAC/scTCR-seq data analysis
  - **Spatial Omics Analysis**: Spatial omics data analysis
  - **Microarray Data Analysis**: SNP and gene expression array analysis
  - **High-Performance Computing**: Slurm and Sun Grid Engine (SGE)
  - **Programming Languages**: R, Python, and Bash
  - **Software and Tools**: Docker, Singularity, Git, LaTeX, SPSS, and ImageJ